

AI-234 Data Structures: Lab 3 Submission

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Objective

To understand the structure of a singly linked list node, dynamically create nodes, and implement standard operations such as insertion, traversal, searching, and deletion in C++.

Theory Concept

A linked list is a dynamic data structure where elements (nodes) are connected using pointers, meaning they are not stored in contiguous memory locations. Each node contains a data field (for the value) and a pointer field (to store the address of the next node).

Lab Tasks Implementation

Task 1: Create a singly linked list containing 5 nodes

Code Snippet:

```
// Created 5 separate nodes in memory
Node* head = new Node();
Node* second = new Node();
Node* third = new Node();
Node* fourth = new Node();
Node* fifth = new Node();

Assign data and link the "next" pointers together
head->data = 10;
head->next = second;
second->data = 20;
second->next = third;
third->data = 30;
third->next = fourth;
fourth->data = 40;
fourth->next = fifth;
fifth->data = 50;

fifth->next = NULL;
```

Task 2: Function to count the number of nodes

Code Snippet:

```
int countNodes(Node* head) {
    int count = 0;
    Node* current = head;
    while (current != NULL) {
        count++;
        current = current->next;
    }
    return count;
}
```

Task 3: Function to insert a node after a given value

Code Snippet:

```
C++
void insertAfter(Node* head, int searchValue, int newValue) {
    Node* current = head;

    while (current != NULL && current->data != searchValue) {
        current = current->next;
    }

    if (current != NULL) {
        Node* newNode = new Node();
        newNode->data = newValue;
        newNode->next = current->next;
        current->next = newNode;
    } else {
        cout << searchValue << " not found in the list." << endl;
    }
}
```

Task 4: Function to delete the last node

Code Snippet:

```
void deleteLast(Node*& head) {
    if (head == NULL) { // List is empty
        cout << "List is empty" << endl;
        return;
    }
    if (head->next == NULL) { // Only one node in the list
        delete head;
        head = NULL;
        return;
    }

    Node* current = head;
    while (current->next->next != NULL) {
        current = current->next;
    }
}
```

```
    }  
    delete current->next;  
    current->next = NULL;  
}
```

Task 5 (Challenge): Complete Menu-Driven Program

Integrated program for all lab manual functions (insert at beginning, insert at end, display, search, delete first)

Complete Source Code:

```
#include <iostream>  
using namespace std;  
  
class Node {  
public:  
    int data;  
    Node* next;  
};  
  
void insertAtEnd(Node*& head, int value) {  
    Node* newNode = new Node();  
    newNode->data = value;  
    newNode->next = NULL;  
    if(head == NULL) {  
        head = newNode;  
        return;  
    }  
    Node* current = head;  
    while(current->next != NULL) {  
        current = current->next;  
    }  
    current->next = newNode;  
}  
  
void insertAtBeginning(Node*& head, int value) {  
    Node* newNode = new Node();  
    newNode->data = value;  
    newNode->next = head;  
    head = newNode;  
}  
  
void display(Node* head) {  
    Node* current = head;  
    while(current != NULL) {  
        cout << current->data << " -> ";  
        current = current->next;  
    }  
    cout << "NULL" << endl;  
}
```

```

bool search(Node* head, int key) {
    Node* current = head;
    while(current != NULL) {
        if(current->data == key)
            return true;
        current = current->next;
    }
    return false;
}

void deleteFirst(Node*& head) {
    if(head == NULL) {
        cout << "List is empty" << endl;
        return;
    }
    Node* temp = head;
    head = head->next;
    delete temp;
    cout << "First node deleted." << endl;
}

int main() {
    Node* head = NULL;
    int choice, value;

    do {
        cout << "\n--- Linked List Menu ---" << endl;
        cout << "1. Insert at Beginning" << endl;
        cout << "2. Insert at End" << endl;
        cout << "3. Display List" << endl;
        cout << "4. Search Element" << endl;
        cout << "5. Delete First Node" << endl;
        cout << "6. Exit" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        switch(choice) {
            case 1:
                cout << "Enter value to insert at beginning: ";
                cin >> value;
                insertAtBeginning(head, value);
                break;
            case 2:
                cout << "Enter value to insert at end: ";
                cin >> value;
                insertAtEnd(head, value);
                break;
            case 3:
                cout << "Current List: ";
                display(head);
                break;
            case 4:
                cout << "Enter element to search: ";

```

```

        cin >> value;
        if(search(head, value))
            cout << value << " Found in the list." << endl;
        else
            cout << value << " Not Found." << endl;
        break;
    case 5:
        deleteFirst(head);
        break;
    case 6:
        cout << "Exiting program..." << endl;
        break;
    default:
        cout << "Invalid choice! Please try again." << endl;
    }
} while(choice != 6);

return 0;
}

```

Output Screenshots:

1. Insert at End:

```

PS C:\Users\User> cd "u:\GITHUB projects\UET-Core-Programming\Semester_03\DSA_LAB\Lab 3\linked_lists" ; if ($?) { g++ LAB3_TASK5.cpp -o LAB3_TASK5 } ; if ($?) { . \LAB3_TASK5 }

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 2
Enter value to insert at end: 10

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 2
Enter value to insert at end: 20

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 3
Current List: 10 -> 20 -> NULL

--- Linked List Menu ---

```

2. Display List:

```
PS C:\Users\User> cd "u:\GITHUB projects\UET-Core-Programming\Semester_03\DSA_LAB\Lab 3\linked_lists\" ; if ($?) { g++ LAB3_TASK5.cpp -o LAB3_TASK5 } ; if ($?) { .\LAB3_TASK5 }
```

```
--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 2
Enter value to insert at end: 10

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 2
Enter value to insert at end: 20

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 3
Current List: 10 -> 20 -> NULL

--- Linked List Menu ---
```

Ln 12, Col 32 Spaces: 4 UTF-8 CRLF {} C++ Go Live Win32

3. Insert at Beginning:

```
U: > GITHUB projects > UET-Core-Programming > Semester_03 > DSA_LAB > Lab 3 > linked_lists > LAB3_TASK5.cpp > insertAtEnd(Node *&, int)
```

```
114 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
```

```
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 3
Current List: 10 -> 20 -> NULL

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 1
Enter value to insert at beginning: 5

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 3
Current List: 5 -> 10 -> 20 -> NULL

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 
```

Restart Visual Studio Code to apply the latest update
Update Now

Ln 12, Col 32 Spaces: 4 UTF-8 CRLF {} C++ Go

4. Search Element:

```
U: > GITHUB projects > UET-Core-Programming > Semester_03 > DSA_LAB > Lab 3 > linked_lists > LAB3_TASK5.cpp > insertAtEnd(Node *8, int)
114
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
6. Exit
Enter your choice: 3
Current List: 5 -> 10 -> 20 -> NULL

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 4
Enter element to search: 20
20 Found in the list.

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 4
Enter element to search: 99
99 Not Found.

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice:

Ln 12, Col 32 Spaces: 4 UTF-8 CRLF {} C++
Update is ready, click to restart. Update Now
```

5. Delete First Node:

```
U: > GITHUB projects > UET-Core-Programming > Semester_03 > DSA_LAB > Lab 3 > linked_lists > LAB3_TASK5.cpp > insertAtEnd(Node *8, int)
114
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
5. Delete First Node
6. Exit
Enter your choice: 4
Enter element to search: 99
99 Not Found.

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 5
First node deleted.

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 3
Current List: 10 -> 20 -> NULL

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice:

Ln 12, Col 32 Spaces: 4 UTF-8 CRLF {} C++
Update is ready, click to restart.
```

6. Exit:

```
U: > GITHUB projects > UET-Core-Programming > Semester_03 > DSA_LAB > Lab 3 > linked_lists > G+ LAB3_TASK5.cpp > insertAtEnd(Node *&, int)
114 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code + - - - - -

Enter your choice: 4
Enter element to search: 99
99 Not Found.

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 5
First node deleted.

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 3
Current List: 10 -> 20 -> NULL

--- Linked List Menu ---
1. Insert at Beginning
2. Insert at End
3. Display List
4. Search Element
5. Delete First Node
6. Exit
Enter your choice: 6
Exiting program...

PS C:\GITHUB\projects\UET-Core-Programming\Semester_03\DSA_LAB\Lab 3\linked_lists>
```

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